

Prove each of the statements in 10-18 by mathematical induction.

5.2.11. $1^3 + 2^3 + \dots + n^3 = \left[\frac{n(n+1)}{2} \right]^2$, for every integer $n \geq 1$

$$1^3 + 2^3 + \dots + n^3 = \sum_{k=1}^n k^3$$

$$\sum_{k=1}^n k^3 = \left[\frac{n(n+1)}{2} \right]^2 = P(n)$$

Proof:

Step 1 (basis step):

We must show that $P(a)$ is true for both sides.

$$a = 1$$

$$\sum_{k=1}^1 k^3 = (1)^3 = 1$$

$$\left[\frac{1(1+1)}{2} \right]^2 = \left[\frac{1(2)}{2} \right]^2 = \left[\frac{2}{2} \right]^2 = [1]^2 = 1$$

Therefore, $P(1)$ is true.

Step 2 (inductive step):

Suppose that $P(k)$ is true, where k is any particular but arbitrarily chosen integer with $k \geq a$.

$$\sum_{i=1}^k i^3 = \left[\frac{k(k+1)}{2} \right]^2$$

We must show that $P(k+1)$ is true for both sides.

We know that for k :

$$\sum_{i=1}^k i^3 = 1^3 + 2^3 + \dots + k^3 = \left[\frac{k(k+1)}{2} \right]^2$$

So, $k+1$ must be:

$$\sum_{i=1}^{k+1} i^3 = 1^3 + 2^3 + \dots + k^3 + (k+1)^3 = \left[\frac{(k+1)(k+2)}{2} \right]^2$$

Since $\sum_{i=1}^{k+1} i^3 = 1^3 + 2^3 + \dots + k^3 + (k+1)^3$ and

$\sum_{i=1}^k i^3 = 1^3 + 2^3 + \dots + k^3$, we can write the left side as:

$$\sum_{i=1}^{k+1} i^3 = \sum_{i=1}^k i^3 + (k+1)^3$$

And now we must show the right hand, where we are looking for the form:

$$\left[\frac{(k+1)(k+2)}{2} \right]^2$$

And since we know that the left side can be expressed

as $\sum_{i=1}^{k+1} i^3 = \sum_{i=1}^k i^3 + (k+1)^3$, we can express the right hand as:

$$\left[\frac{k(k+1)}{2} \right]^2 + (k+1)^3$$

Using algebra:

$$\left[\frac{k(k+1)}{2} \right]^2 = \frac{k^2(k+1)^2}{4}$$

$$(k+1)^3 = (k+1)^2(k+1)$$

$$\frac{k^2(k+1)^2}{4} + (k+1)^2(k+1)$$

$$(k+1)^2 \left[\frac{k^2}{4} + (k+1) \right]$$

$$\frac{k^2}{4} + (k+1) = \frac{k^2 + 4k + 4}{4}$$

$$(k+1)^2 \left[\frac{k^2 + 4k + 4}{4} \right]$$

$$\frac{k^2 + 4k + 4}{4} = \frac{(k+2)^2}{4}$$

$$(k+1)^2 \left[\frac{(k+2)^2}{4} \right]$$

$$\frac{(k+1)^2(k+2)^2}{4} = \left[\frac{(k+1)(k+2)}{2} \right]^2$$

Therefore, $P(k+1)$ is true for both sides, which is what was to be shown.

□

Solution: $1^3 + 2^3 + \dots + n^3 = \left[\frac{n(n+1)}{2} \right]^2$, for every integer $n \geq 1$ is true.